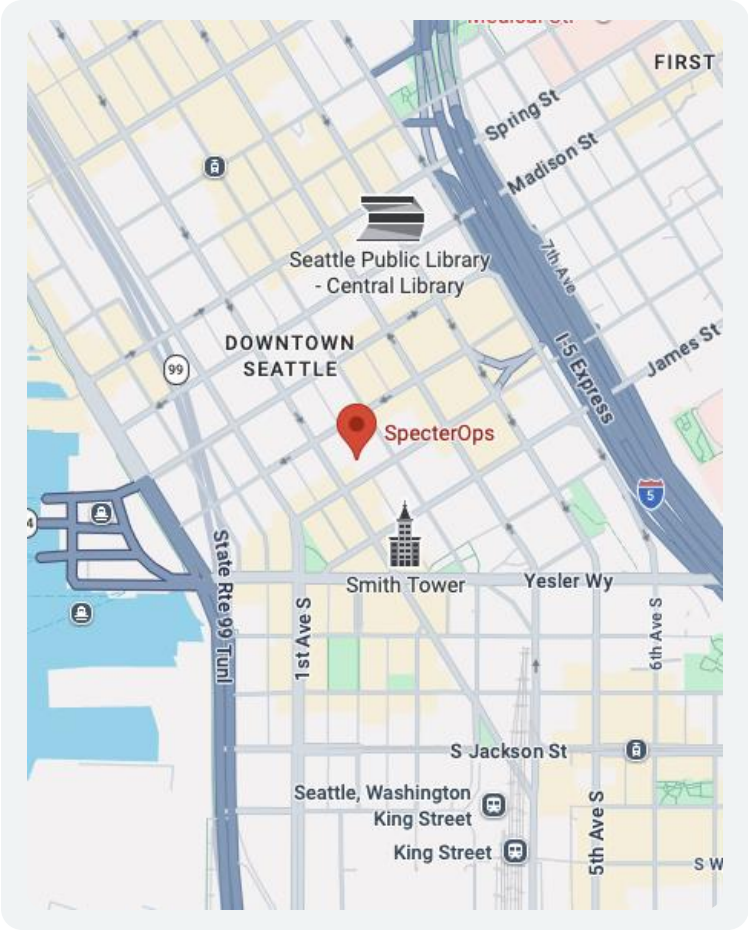
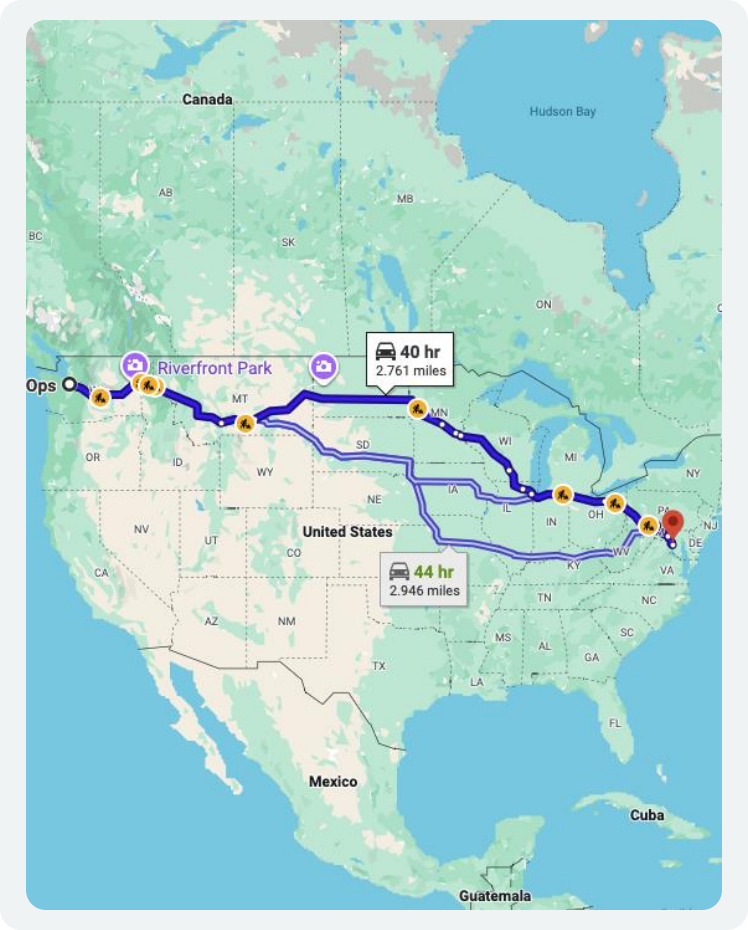
The Identity APM Maturity Model

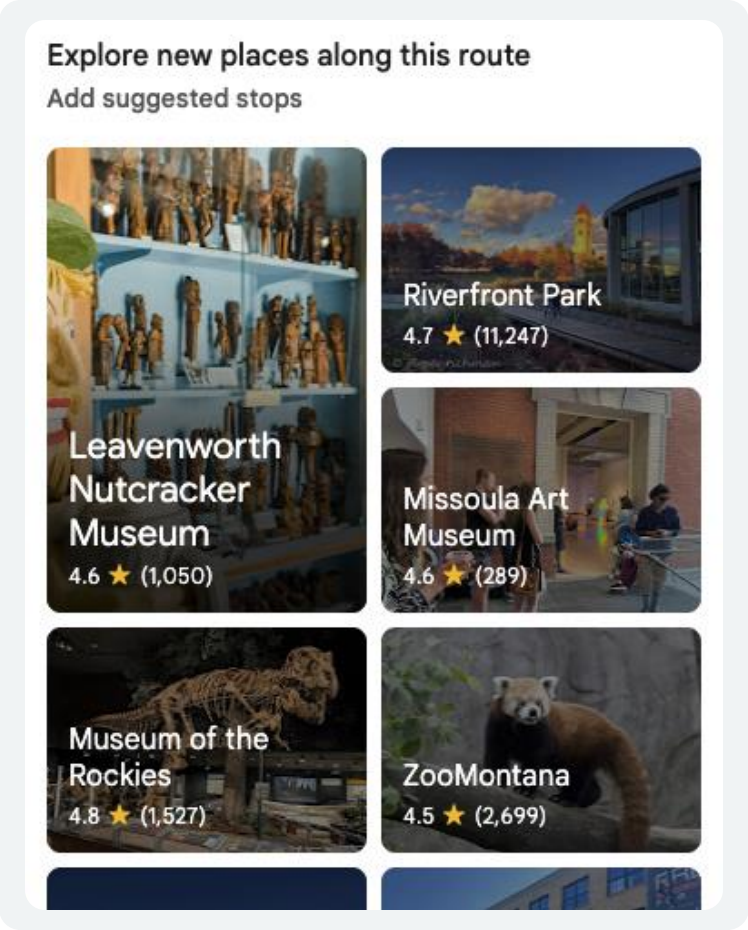
A Framework for Leveling Up Your APM Practice

Kate Dawson, Nathan Davis & Hugo van den Toorn
SpecterOps

The Trusted Analogy







APM JOURNEY

YOUR TOUR GUIDES

On this APM Journey

Kate Dawson

Voice of Reason

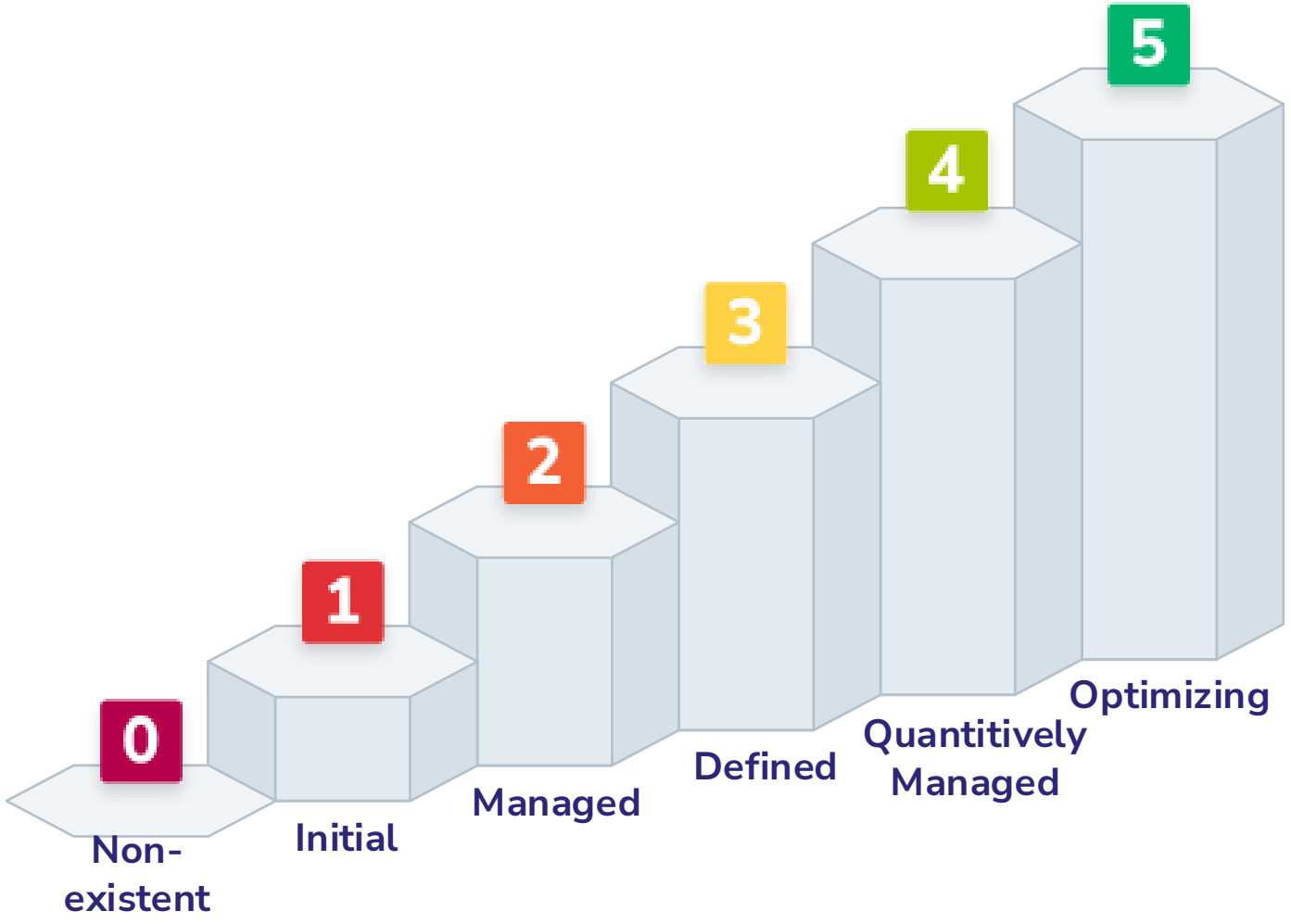
Nathan Davis

Voice of Tech

Hugo van den Toorn

Voice of the CISO

Six Levels of Maturity



Three Dimensions

PEOPLE

Tools alone don't remediate attack paths.

PROCESS

Without repeatable processes, APM becomes reactive firefighting.

TECHNOLOGY

Visibility alone does not solve problems.

Initial Maturity Model

https://ghst.ly/maturitymodel

Initial Maturity Model

Fourteen Aspects

PEOPLE

- PEO 1
- PEO 2
- PEO 3
- PEO 4

PROCESS

- PRO 1
- PRO 2
- PRO 3
- PRO 4
- PRO 5

TECHNOLOGY

- TEC 1
- TEC 2
- TEC 3
- TEC 4
- TEC 5

People

PEO 1

Executive Support & Resources

Level of corporate buy-in, dedicated budget and headcount.

PEO 2

Cross-Functional Coordination

Coordination and effectiveness of collaboration between departments.

PEO 3

Roles & Responsibilities

Level of APM ownership and accountability in the organization.

PEO 4

Staffing & Skills

Formality of training and skills specific to APM.

Process

PRO 1

Discover Frequency

Frequency and speed of attack path discovery.

PRO 2

Mean Time to Remediate

Average time to remediate attack paths.

PRO 3

Validation & Regression Testing

Prevention of regression and tracking thereof.

PRO 4

Process Integration

Level of integration in business processes across the organization.

PRO 5

Documentation & Playbooks

Standardization of attack path remediation or prevention.

Technology

01
10

TEC 1

Environment Coverage

Percentage of coverage of identity provider(s).

TEC 2

Tool Integration Depth

Level of integration of APM within tech-stack.

TEC 3

Path Visibility Capability

Types of Adversary Tradecraft covered by APM capabilities.

TEC 4

Automation Level

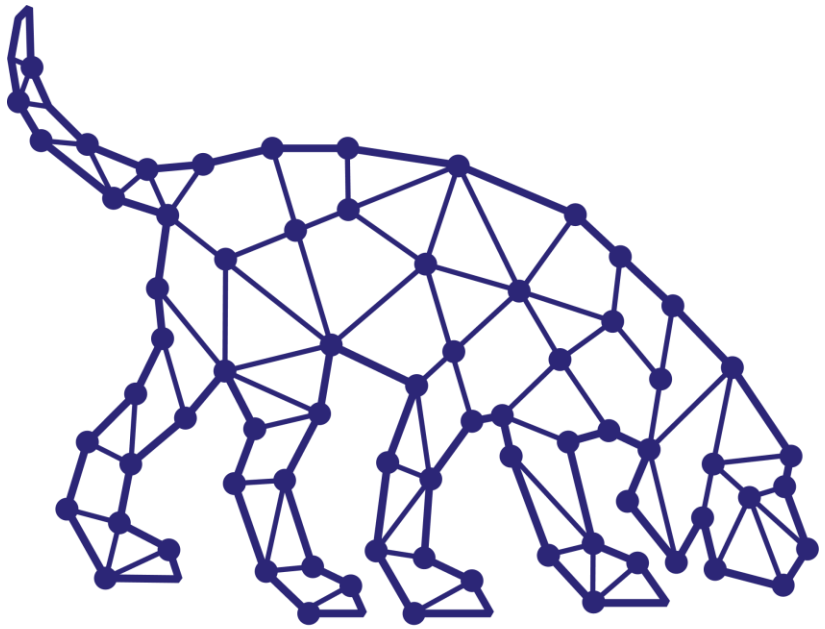
APM automation, ranging from manual to fully automated.

TEC 5

Risk Quantification

From subjective, to broadly-quantitative, to specifically-quantified, to predictive.

Agnostic Model



BLOODHOUND
ENTERPRISE

MATURITY MODEL

DIMENSION	ASPECT		MATURITY LEVEL
PEOPLE	PEO 1	Executive Support & Resources	
	PEO 2	Cross-Functional Coordination	
	PEO 3	Roles & Responsibilities	
	PEO 4	Staffing & Skills	
PROCESS	PEO 1	Discover Frequency	
	PEO 2	Mean Time to Remediate	
	PEO 3	Validation & Regression	
	PEO 4	Process Integration	
	PEO 5	Documentation & Playbooks	
TECHNOLOGY	PRO 1	Environment Coverage	
	PRO 2	Tool Integration Depth	
	PRO 3	Path Visibility Capability	
	PRO 4	Automation Level	
	PRO 5	Risk Quantification	

MATURITY MODEL

DIMENSION	ASPECT		Kate	Nathan	Hugo
					STORY
PEOPLE	PEO 1	Executive Support & Resources			Costly Red Teaming vs a tool
	PEO 2	Cross-Functional Coordination			”BloodHound is scary”
	PEO 3	Roles & Responsibilities			Who's on first?
	PEO 4	Staffing & Skills			Ask for help! No unique finding
PROCESS	PEO 1	Discover Frequency			Automation is your friend
	PEO 2	Mean Time to Remediate			Know thyself
	PEO 3	Validation & Regression			ADCS... Again
	PEO 4	Process Integration			“Just login once a month”
	PEO 5	Documentation & Playbooks			
TECHNOLOGY	PRO 1	Environment Coverage			(Open)Graph all the things
	PRO 2	Tool Integration Depth			Privileged collection
	PRO 3	Path Visibility Capability			No attack paths != no problem
	PRO 4	Automation Level			Tines Ticket Enrichment
	PRO 5	Risk Quantification			Beyond Tier Zero

MATURITY MODEL

DIMENSION	ASPECT		
PEOPLE	PEO 1	Executive Support & Resources	Sparking interest & Gaining necessary knowledge EXCITE
	PEO 2	Cross-Functional Coordination	
	PEO 3	Roles & Responsibilities	
	PEO 4	Staffing & Skills	
PROCESS	PEO 1	Discover Frequency	A new security program, similar to Vulnerability Management. BUILD
	PEO 2	Mean Time to Remediate	
	PEO 3	Validation & Regression	
	PEO 4	Process Integration	
	PEO 5	Documentation & Playbooks	
TECHNOLOGY	PRO 1	Environment Coverage	Your Tech stack, APM tools and extensions INTEGRATE
	PRO 2	Tool Integration Depth	
	PRO 3	Path Visibility Capability	
	PRO 4	Automation Level	
	PRO 5	Risk Quantification	

To get you started on your APM journey

Guidance: CISO's guide

https://ghst.ly/cisoguide

Direction: Maturity Model

https://ghst.ly/maturitymodel

Action: (Self-)Assessment

Do the assessment or talk to your TAM

https://ghst.ly/apmmaturitytool

Thank you!
Please share your feedback